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Inventor(s)	Cheney; Craig et al.

Rapid-entry footwear having a transforming footbed

Abstract

A rapid-entry shoe with a sole portion and a footbed that slide, pivot or translate relative to each other to provide both an open configuration (for ease of donning or doffing by a foot) and a closed configuration (for securement of a foot).

Inventors: Cheney; Craig (Lindon, UT), Munger; Seth (Springville, UT), Johnson; Bradley (Draper, UT)

Applicant: FAST IP, LLC (Lindon, UT)

Family ID: 1000008820933

Assignee: FAST IP, LLC (Lindon, UT)

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Primary Examiner: Bays; Marie D

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of, claims priority to and the benefit of U.S. Ser. No. 17/827,954 filed May 30, 2022 and entitled "RAPID-ENTRY FOOTWEAR HAVING A TRANSFORMING FOOTBED." U.S. Ser. No. 17/827,954 is a continuation of, claims priority to and the benefit of PCT Serial No. PCT/US22/16867 filed Feb. 17, 2022 and entitled "RAPID-ENTRY FOOTWEAR HAVING A TRANSFORMING FOOTBED." PCT Serial No. PCT/US22/16867 claims the benefit of U.S. Provisional Patent Application No. 63/150,550, filed Feb. 17, 2021 and entitled "RAPID-ENTRY FOOTWEAR HAVING A TRANSLATING FOOTBED." All of the aforementioned applications are incorporated herein by reference in their entireties.

FIELD

(1) The present disclosure relates to footwear, and more particularly to rapid-entry footwear having a transforming footbed.

BACKGROUND

(2) Whether due to inconvenience or inability, donning and doffing of shoes, including tying or otherwise securing the same, may be undesirable and/or present difficulties to some individuals. The present disclosure addresses this need.

SUMMARY

(3) In example embodiments, a rapid-entry shoe of the present disclosure comprises a sole portion, a footbed and a separable upper. In accordance with an aspect of the present disclosure, the footbed is configured to transform (e.g., slide or pivot or translate or bow) with a first portion of the separable upper relative to the sole portion and a second portion of the separable upper between an open configuration having an expanded upper opening for ease of donning or doffing by a foot and a closed configuration having an unexpanded upper opening for securement of the foot. In accordance with an aspect of the present disclosure, in the closed configuration, the footbed is secured against the sole portion.

(4) A rapid-entry shoe in accordance with the present disclosure comprises a sole portion, a footbed, and an upper. In accordance with example embodiments, the upper comprises an upper forward portion and an upper heel portion. In accordance with example embodiments, the sole portion is coupled to the upper heel portion.

(5) In accordance with example embodiments, the footbed is coupled to the upper forward portion. In accordance with example embodiments, the footbed is configured to pivot relative to the sole portion between an open configuration, in which the footbed is angled relative to the sole portion,

and a closed position, in which the footbed and the sole portion are substantially parallel to one another.

(6) In accordance with example embodiments, the footbed is coupled to the upper forward portion to form a bootie. In accordance with example embodiments, the bootie is configured to pivot relative to the sole portion between an open configuration, in which a rear edge of the footbed is not in contact with a rear edge of the sole portion, and a closed position, in which the rear edge of the footbed is in contact with the rear edge of the sole portion.

(7) In accordance with example embodiments, the bootie comprises a plurality of laces to secure a foot within the bootie.

(8) In accordance with example embodiments, in the closed configuration, the rear edge of the footbed is positioned underneath at least one bumper coupled to an inner surface of the upper heel portion, the at least one bumper configured to secure the bootie in the closed position. In accordance with example embodiments, the at least one bumper is comprised of a foam material. In accordance with example embodiments, the at least one bumper comprises a dynamic bumper configured to be manually buckled such that, in the closed position, the rear edge of the footbed is no longer positioned underneath the at least one bumper, so as to no longer secure the bootie in the closed position.

(9) In accordance with example embodiments, in the closed configuration, the rear edge of the footbed is positioned within an aperture of a dynamic hook, the aperture configured to secure the bootie in the closed position. In accordance with example embodiments, the dynamic hook comprises a living hinge configured to be manually opened such that, in the closed position, the rear edge of the footbed is no longer positioned within the aperture, so as to no longer secure the bootie in the closed position.

(10) In accordance with example embodiments, the upper forward portion comprises a foot hold extending completely under the footbed between opposing sides of the footbed. In accordance with example embodiments, the foot hold comprises a plurality of laces.

(11) In accordance with example embodiments, the upper forward portion is coupled to the sole portion exclusively at a forward portion of the sole portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings may provide a further understanding of example embodiments of the present disclosure and are incorporated in, and constitute a part of, this specification. In the accompanying drawings, only one rapid-entry shoe (either a left shoe or a right shoe) may be illustrated, however, it should be understood that in such instances, the illustrated shoe may be mirror-imaged so as to be the other shoe. The use of like reference numerals throughout the accompanying drawings is for convenience only, and should not be construed as implying that any of the illustrated embodiments are equivalent. The accompanying drawings are for purposes of illustration and not of limitation.

(2) FIGS. 1A-1C illustrate footwear with a sliding footbed, in accordance with example embodiments of the present disclosure.

(3) FIGS. 1D and 1E illustrate another embodiment of footwear with a sliding footbed, in accordance with the present disclosure.

(4) FIGS. 2A-2D illustrate footwear with a translating footbed and upper heel portion, in accordance with example embodiments of the present disclosure.

(5) FIGS. 3A-3D illustrate footwear with a pivoting footbed, in accordance with example embodiments of the present disclosure.

(6) FIGS. 3E and 3F illustrate another embodiment of footwear with a pivoting footbed, in

accordance with the present disclosure.

(7) FIGS. 4A-4C illustrate footwear having the profile of a high-top shoe with a pivoting footbed, in accordance with example embodiments of the present disclosure.

(8) FIGS. 5A-5C illustrate footwear with a translating footbed and upper forward portion, in accordance with example embodiments of the present disclosure.

(9) FIGS. 6A-6E illustrate a rapid-entry shoe having a stationary bumper to secure a footbed, in accordance with example embodiments of the present disclosure.

(10) FIGS. 6F-6I illustrate rapid-entry shoes having different configurations of stationary bumpers, in accordance with example embodiments of the present disclosure.

(11) FIGS. 7A and 7B illustrate a rapid-entry shoe having a dynamic bumper to secure a footbed, in accordance with example embodiments of the present disclosure.

(12) FIGS. 8A and 8B illustrate a rapid-entry shoe having a dynamic hook to secure a footbed, in accordance with example embodiments of the present disclosure.

(13) FIGS. 9A and 9B illustrate a rapid-entry shoe having a pivoting upper heel portion and a pivoting upper forward portion, in accordance with example embodiments of the present disclosure.

(14) FIGS. 10A and 10B illustrate, in accordance with example embodiments of the present disclosure.

DETAILED DESCRIPTION

(15) Example embodiments of the present disclosure are described in sufficient detail in this detailed description to enable persons having ordinary skill in the relevant art to practice the present disclosure, however, it should be understood that other embodiments may be realized and that mechanical and chemical changes may be made without departing from the spirit or scope of the present disclosure. Thus, this detailed description is for purposes of illustration and not of limitation.

(16) For example, unless the context dictates otherwise, example embodiments described herein may be combined with other embodiments described herein. Similarly, references to “example embodiment,” “example embodiments” and the like indicate that the embodiment(s) described may comprise a particular feature, structure, or characteristic, but every embodiment may not necessarily comprise the particular feature, structure, or characteristic. Moreover, such references may not necessarily refer to the same embodiment(s). Any reference to singular includes plural embodiments, and any reference to plural includes singular embodiments.

(17) Any reference to coupled, connected, attached or the like may be temporary or permanent, removeable or not, non-integral or integral, partial or full, and may be facilitated by one or more of adhesives, stitches, hook and loop fasteners, buttons, clips, grommets, zippers and other means known in the art or hereinafter developed.

(18) As used herein, the transitional term “comprising”, which is synonymous with “including,” “containing,” or “characterized by,” is inclusive or open-ended and does not exclude additional, unrecited elements or method steps. The transitional phrase “consisting of” excludes any element, step, or ingredient not specified in the claim. The transitional phrase “consisting essentially of” limits the scope of a claim to the specified materials or steps “and those that do not materially affect the basic and novel characteristic(s)” of the claimed invention.

(19) No claim limitation is intended to invoke 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph or the like unless it explicitly uses the term “means” and includes functional language.

(20) In describing example embodiments of the rapid-entry footwear, certain directional terms may be used. By way of example, terms such as “right,” “left,” “medial,” “lateral,” “front,” “back,” “forward,” “backward,” “rearward,” “top,” “bottom,” “upper,” “lower,” “up,” “down,” and the like may be used to describe example embodiments of the rapid-entry footwear. These terms should be given meaning according to the manner in which the rapid-entry footwear is most typically designed for use, with the rapid-entry footwear on a user's foot and with the user's shod foot

disposed on or ready for placement on an underlying surface. Thus, these directions may be understood relative to the rapid-entry footwear in such use. Similarly, as the rapid-entry footwear is intended primarily for use as footwear, terms such as “inner,” “inward,” “outer,” “outward,” “innermost,” “outermost,” “inside,” “outside,” and the like should be understood in reference to the rapid-entry footwear's intended use, such that inner, inward, innermost, inside, and the like signify relatively closer to the user's foot, and outer, outward, outermost, outside, and the like signify relatively farther from the user's foot when the rapid-entry footwear is being used for its intended purpose. Notwithstanding the foregoing, if the foregoing definitional guidance is contradicted by an individual use herein of any of the foregoing terms, the term should be understood and read according to the definition that gives life and meaning to the particular instance of the term.

(21) As used herein, unless the context dictates otherwise, a “rapid-entry shoe” refers to an athleisure shoe, a casual shoe, a formal shoe, a dress shoe, a heel, a sports/athletic shoe (e.g., a tennis shoe, a golf shoe, a bowling shoe, a running shoe, a basketball shoe, a soccer shoe, a ballet shoe, etc.), a walking shoe, a sandal, a boot, or other suitable type of shoe. Additionally, a rapid-entry shoe can be sized and configured to be worn by men, women, or children.

(22) As used herein, unless the context dictates otherwise, “footbed” can refer to an insole, sock liner, wedge or midsole, and “sole portion” can refer to a midsole or outsole.

(23) As used herein, unless the context dictates otherwise, “upper heel portion” comprises a portion of the upper extending between medial and lateral sides, around the heel (e.g., heel counter and quarters), and “upper forward portion” can refer to comprises a portion of the upper extending between medial and lateral sides, over the forefoot (e.g., tongue, toe cap, vamp and quarters).

(24) As used herein, unless the context dictates otherwise, “separable” refers to having at least two parts capable of relative movement, excluding a tongue in the case of a separable upper. As used herein, unless the context dictates otherwise, “slide” refers to relative 1-dimensional movement (e.g., along an axis parallel to the length of the footbed) while remaining substantially parallel. As used herein, unless the context dictates otherwise, “pivot” refers to relative movement resulting in an angle. As used herein, unless the context dictates otherwise, “translate” refers to relative 2-dimensional movement (e.g., along axes parallel and orthogonal to the length of the footbed) while remaining substantially parallel.

(25) In example embodiments of the present disclosure, a footbed of a rapid-entry shoe is configured to move relative to a sole portion of a rapid-entry shoe, between an open configuration (for ease of donning or doffing by a foot) and a closed configuration (for securement of a foot). In example embodiments of the present disclosure, a foot applying a force (e.g., a downward and/or shear force) to the footbed moves it relative to the sole portion from an open configuration to a closed configuration. In connection with example embodiments, a sole portion of the rapid-entry shoe can be configured to remain substantially planar or otherwise not bend during the transition of a footbed of the rapid-entry shoe between the open and closed configurations.

(26) In some embodiments, and with reference to FIGS. 1A-2D, an upper of a rapid-entry shoe **100** is a separable upper. In this regard, the upper can comprise an upper heel portion **130** and an upper forward portion **120**. In example embodiments, an upper heel portion **130** of a rapid-entry shoe **100** is coupled to a footbed **140** and configured to slide, pivot or translate between an open configuration and a closed configuration relative to a sole portion **110** coupled to an upper forward portion **120**. In connection with the foregoing embodiments, a bootie **160** can be coupled to the footbed **140**. The bootie **160** can extend the length of the footbed (e.g., FIG. 1A) or can cover only a forward portion of the footbed and be open on the sides (e.g., FIG. 2A).

(27) With reference now to FIGS. 1A-1E, in such embodiments, all or a portion of a footbed **140** can be configured to slide with an upper heel portion **130** rearward relative to a sole portion **110** and an upper forward portion **120**. In such embodiments, the footbed **140** and the sole portion **110** can remain parallel or substantially parallel during the sliding. In such embodiments, sliding can be facilitated by corresponding male and female track components on the footbed **140** and the sole

portion **110** or the upper forward portion **120**. The upper heel portion **130** and the upper forward portion **120** can be secured to each other by a securement element (e.g., ratchet, strap) **136** comprising a securement release/tether **136**.

(28) Paying particular attention to FIGS. **1D** and **1E**, an upper heel portion **130** and an upper forward portion **120** can be coupled to each other on one or both sides by a gore **150**. The gore **150** may enhance weatherproofing of the rapid-entry shoe **100**. The gore **150**, in turn, can comprise a stretch material and/or a baffle material. In this regard, the upper heel portion **130** can be spring-loaded so as to be biased with the coupled footbed in the open configuration (e.g., FIG. **1D**).

(29) With reference to FIGS. **2A-2D**, in such embodiments, all or a portion of a footbed **140** can be configured to translate with an upper heel portion **130** upward and rearward relative to a sole portion **110** and an upper forward portion **120**. In such embodiments, the footbed **140** and the sole portion **110** can remain parallel or substantially parallel during the translation. In such embodiments, translation can be facilitated by a lever arm **142** extending at an angle between the sole portion **110** and a top (FIGS. **2C** and **2D**), middle, or bottom (FIGS. **2A** and **2B**) of the upper heel portion **130** (or between the sole portion **110** and the footbed **140**), the lever arm **142** being configured to rotate (e.g., a circumferential, pin or ball bearing) relative to one or both of the footbed **140** and the sole portion **110**.

(30) A lever arm **142** can be on a single side of the rapid-entry shoe or on both sides of the rapid-entry shoe. A lever arm **142** can be coupled to the sole portion rearward relative to where the lever arm is coupled to the upper heel portion when in the closed configuration but forward relative to where the lever arm is coupled to the upper heel portion when in the open configuration. In such embodiments, the lever arm **142** can lock by going past center.

(31) Rather than being coupled to a side of a sole portion **110** or footbed **140** as illustrated, a lever arm **142** can be coupled to an upper surface of the sole portion **110** and a lower surface of the footbed **140** and be received with a track within the sole portion **110** and/or the footbed **140**. In such embodiments, a single lever arm **142** can be used (e.g., approximately equidistance between opposing sides of rapid entry-shoe **100**) or a plurality of lever arms **142** can be used (e.g., **2**, on opposing sides of rapid entry-shoe **100**).

(32) In other embodiments, and with reference to FIGS. **3A-5C**, an upper of a rapid-entry shoe **100** is a separable upper. In this regard, the upper can comprise an upper heel portion **130** and an upper forward portion **120**. In example embodiments, an upper forward portion **120** of a rapid-entry shoe **100** is coupled to a footbed **140** and configured to slide, pivot or translate between an open configuration and a closed configuration relative to a sole portion **110** coupled to an upper heel portion **130**. In example embodiments, the upper forward portion **120** is not merely the tongue of the rapid-entry shoe, but instead, comprises the complete upper (excluding the upper heel portion **130**). In this regard, the upper forward portion **120** can be coupled to the sole portion **110** exclusively at a forward portion of the sole portion (e.g., at or proximal to the site of the pin hinge **144** in FIGS. **3E** and **3F**). Stated another way, in example embodiments, the upper forward portion **120** is not coupled to the sole portion **110** at a rearward portion of the sole portion.

(33) In example embodiments, a foot hold **170** can be coupled to the footbed **140**. The foot hold **170** can comprise one or more of a strap (e.g., FIG. **3A**), one or a plurality of laces (e.g., FIGS. **4A** and **5A**) and a bootie (e.g., FIGS. **3A**, **4A**, **5A**). In example embodiments, the foot hold **170** terminates at and is coupled exclusively to opposing sides of the footbed **140**, while in other embodiments, the foot hold **170** extends completely under the footbed **140** between opposing sides of the footbed **140**. In example embodiments, unlike with conventional shoes, one or a plurality of laces are coupled to the upper and have vertices that are moveable relative to the sole portion.

(34) With reference now to FIGS. **3A-4C**, in such embodiments, all or a portion of a footbed **140** can be configured to pivot with an upper forward portion **120** upward relative to a sole portion **110** and an upper heel portion **130**. In some embodiments, the sole portion **110** is configured to arc or bend during the pivoting. In other embodiments, the sole portion **110** is configured to remain

substantially planar or otherwise not bend during the pivoting. In example embodiments, pivoting can be facilitated by a hinge (e.g., a living hinge or a pin hinge) between the footbed **140** and the sole portion **110**. Pivoting can also be facilitated by the footbed **140** being resiliently deformable but stable in (e.g., biased toward) a non-planar or otherwise bent shape. In example embodiments, the footbed **140** can be secured into a rapid-entry shoe **100** against the sole portion **110** with a stationary bumper **112**, discussed below.

(35) Paying particular attention to FIGS. 3E and 3F, an upper heel portion **130** and an upper forward portion **120** can be coupled to each other on one or both sides by a gusset **155**. The gusset **155** may enhance weatherproofing of the rapid-entry shoe **100**. The gusset **155**, in turn, can comprise a stretch material and/or a baffle material. The upper heel portion **130** and the upper forward portion **120** can be secured to each another by a securement element (e.g., ratchet, strap) **136** comprising a securement release/tether **136**.

(36) With continued attention to FIGS. 3E and 3F, an upper heel portion **130** (and a sole portion **110**) and an upper forward portion **120** (and a footbed **140**) can be configured to pivot relative to each other via a pin hinge **144** (e.g., a spring-loaded pin hinge **144**), a first side of the pin hinge **144** coupled to the upper heel portion **130** and a second side of the pin hinge **144** coupled to the upper forward portion **120**.

(37) Turning now to FIGS. 4A-4C, a rapid-entry shoe **100** can comprise an upper heel portion **130** and an upper forward portion **120** both having the profile of a high-top shoe (e.g., a basketball shoe or a boot). In example embodiments, an additional strap or band may added over the top of the upper, coupled to and between opposing quarters, bluchers or upper panels, to secure the same together and/or secure the upper forward portion **120** and the footbed **140** relative to the sole portion **110** and the upper heel portion **130**.

(38) With reference to FIGS. 5A-5C, in such embodiments, all or a portion of a footbed **140** can be configured to translate with an upper forward portion **120** upward and forward relative to a sole portion **110** and an upper heel portion **130**. In such embodiments, the footbed **140** and the sole portion **110** can remain parallel or substantially parallel during the translation. In such embodiments, translation can be facilitated by a lever arm **142** extending at an angle between the sole portion **110** and the upper forward portion **120** (or between the sole portion **110** and the footbed **140**), the lever arm **142** being configured to rotate (e.g., a circumferential, pin or ball bearing) relative to one or both of the footbed **140** and the sole portion **110**.

(39) A lever arm **142** can be on a single side of the rapid-entry shoe or on both sides of the rapid-entry shoe. A lever arm **142** can be coupled to the sole portion forward relative to where the lever arm is coupled to the upper forward portion, when in both an open and a closed configuration.

(40) Rather than being coupled to a side of a sole portion **110** or footbed **140** as illustrated, a lever arm **142** can be coupled to an upper surface of the sole portion **110** and a lower surface of the footbed **140** and be received with a track within the sole portion **110** and/or the footbed **140**. In such embodiments, a single lever arm **142** can be used (e.g., approximately equidistance between opposing sides of rapid entry-shoe **100**) or a plurality of lever arms **142** can be used (e.g., 2, on opposing sides of rapid entry-shoe **100**).

(41) In example embodiments, and with continued reference to FIGS. 5A-5C, a rapid-entry shoe **100** is secured in a closed configuration with one or more securement elements **146** comprising one or more of a hook and loop fastener, magnet, clip, tie, lever, ratchet or the like.

(42) Alternatively, in accordance with any of the example embodiments described herein (e.g., FIGS. 3A-5C), when in the closed configuration, a footbed can be secured into a rapid-entry shoe against the sole portion with one or more of a stationary bumper, a dynamic bumper and a dynamic hook.

(43) With reference to FIGS. 6A-6E, a rapid-entry shoe **100** can comprise a “bootie construction” that connects a tongue and a footbed **140** so they can move independently from the rest of the rapid-entry shoe **100**. In such embodiments, a footbed **140** can be secured into a rapid-entry shoe

100 against the sole portion **110** with a stationary bumper **112** having a lip at its bottom (i.e., the stationary bumper **112** does not extend completely to the sole portion **110**), the lip defining an aperture for receiving and securing an end of the footbed **140** underneath the lip. A foot applying a force (e.g., a downward and/or shear force) to the footbed **140** pushes an end of the footbed **140** past the lip and locks it under the lip, within the aperture. In this regard, an upper surface of the end of the footbed **140** be more rigid than a lower surface of the end of the footbed **140** to facilitate movement of the end of the footbed **140** past the lip and retention of it under the lip. Alternatively, a lower surface of the end of the footbed **140** can comprise a skid plate to facilitate movement of the end of the footbed **140** past the lip. In some embodiments, the footbed **140** can be compressible along an axis parallel to the length of the footbed.

(44) Additionally, with specific reference to FIG. 6D, articulation of the sole portion during motion results in the footbed **140** assuming an inverted arching shape that shears the footbed **140** rearward, pushing it tighter within the aperture and under the lip, locking the footbed **140** in tighter. In example embodiments, the stationary bumper **112** is configured to be manually buckled (e.g., with a toe of the other foot) such that, in the closed position, the rear edge or end of the footbed **140** is no longer positioned underneath the lip of the stationary bumper **112**, so as to no longer secure the footbed **140** in the locked or closed position.

(45) With reference to FIGS. 6F-6I, a footbed **140** can be secured into a rapid-entry shoe **100** against the sole portion **110** with a stationary bumper **112** at a rear of the upper heel portion (FIG. 6F), at a rear and opposing sides of the upper heel portion (FIG. 6G), at a rear and a single side of the upper heel portion (FIG. 6H) or at opposing sides of the upper heel portion (FIG. 6I). A stationary bumper **112** can be comprised of foam.

(46) With reference to FIGS. 7A and 7B, a footbed **140** can be secured into a rapid-entry shoe **100** against the sole portion **110** with a dynamic bumper **114** having a lip at its bottom (i.e., the dynamic bumper **114** does not extend completely to the sole portion **110**), the lip moveable from a position above an end of the footbed **140** (i.e., the footbed **140** locked down, FIG. 7A) to a position not above an end of the footbed **140** (i.e., the footbed **140** no longer locked down, FIG. 7B). In this regard, the dynamic bumper **114** can comprise a bistable elongated element or plate (e.g., bistable around an actuation point). In example embodiments, the dynamic bumper **114** is configured to be manually buckled (e.g., with a toe of the other foot) such that, in the closed position, the rear edge or end of the footbed **140** is no longer positioned underneath the lip of the dynamic bumper **114**, so as to no longer secure the footbed **140** in the locked or closed position.

(47) With reference to FIGS. 8A and 8B, a footbed **140** can be secured into a rapid-entry shoe **100** against the sole portion **110** with a dynamic hook **116**, the dynamic hook **116** moveable from a position above an end of the footbed **140** (i.e., the footbed **140** locked down within an aperture **117**, FIG. 8A) to a position not above an end of the footbed **140** (i.e., the footbed **140** no longer locked down within the aperture **117**, FIG. 8B). In this regard, the dynamic hook **116** can comprise a living hinge. In example embodiments, the dynamic hook **116** is configured to be manually opened such that, in the closed position, the rear edge of the footbed **140** is no longer positioned within the aperture **117**, so as to no longer secure the footbed **140** in the locked or closed position. The dynamic hook **116** can be formed in the foam of the midsole or comprise a TPU heel clip.

(48) With reference to FIGS. 9A and 9B, in still other embodiments, all or a portion of a footbed **140** of a rapid-entry shoe **100** can be configured to bow upward, independent from or together with an upper, away from a sole portion **110**. In such embodiments, an upper heel portion **130** and an upper forward portion **120** can pivot relative to each other, for example, along the upward bow of the footbed **140**.

(49) With reference to FIGS. 10A and 10B, in still other embodiments, all or a portion of a footbed **140** of a rapid-entry shoe **100** can be configured to bow upward, independent from or together with an upper, away from a sole portion **110**. In such embodiments, an upper heel portion **130** can pivot relative to the sole portion **110** and the footbed **140** can be received in a recess **118** in the sole

portion 110.

(50) In accordance with any of the example embodiments described herein, a footbed can be temporarily engaged relative to a sole portion in an open configuration or a closed configuration by a securement element comprising one or more of a hook and loop fastener, magnet, clip, tie, lever, ratchet or the like. In some embodiments, the securement element can be released (i.e., disengaged) by the opposite foot or rapid-entry shoe.

(51) In accordance with any of the example embodiments described herein, a collapsible or stretchable layer can be disposed between a footbed and a sole portion such that it expands in an open configuration and collapses in a closed configuration. In this regard, there can be continuity between the footbed and the sole portion to prevent entry of dirt, debris, liquid, heat, cold or the like.

(52) It will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the spirit or scope of the disclosure. Thus, it is intended that the embodiments described herein cover the modifications and variations of this disclosure provided they come within the scope of the appended claims and their equivalents.

(53) Numerous characteristics and advantages have been set forth in the preceding description, including various alternatives together with details of the structure and function of the devices and/or methods. The disclosure is intended as illustrative only and as such is not intended to be exhaustive. It will be evident to those skilled in the art that various modifications can be made, especially in matters of structure, materials, elements, components, shape, size and arrangement of parts including combinations within the principles of the invention, to the full extent indicated by the broad, general meaning of the terms in which the appended claims are expressed. To the extent that these various modifications do not depart from the spirit and scope of the appended claims, they are intended to be encompassed therein.

Claims

1. A rapid-entry shoe comprising: a sole portion; a footbed; and an upper, the upper comprising an upper forward portion and an upper heel portion; wherein the sole portion is coupled to the upper heel portion; wherein the footbed is coupled to the upper forward portion; wherein the footbed is configured to pivot relative to the sole portion between an open configuration, in which a rear edge of the footbed is not in contact with an upper surface of the sole portion, and a closed position, in which the rear edge of the footbed is in contact with the upper surface of the sole portion; wherein, in the closed configuration, the rear edge of the footbed is positioned underneath at least one bumper coupled to an inner surface of the upper heel portion, the at least one bumper configured to secure the footbed in the closed position; and wherein the sole portion is configured to remain substantially planar or not bend during transition between the open configuration and the closed configuration.

2. The rapid-entry shoe of claim 1, wherein the at least one bumper is comprised of a foam material.

3. The rapid-entry shoe of claim 1, wherein, in the open configuration, a rear portion of the footbed is angled with respect to a forward portion of the footbed.

4. A rapid-entry shoe comprising: a sole portion; a footbed; and an upper, the upper comprising an upper forward portion and an upper heel portion; wherein the sole portion is coupled to the upper heel portion; wherein the footbed is coupled to the upper forward portion to form an upper construction; wherein the upper construction is configured to pivot relative to the sole portion between an open configuration in which a rear portion of the footbed is angled with respect to a rear portion of the sole portion, and a closed position in which both the rear portion of the footbed is adjacent to the sole portion; wherein, in the closed position, a rear edge of the footbed is positioned underneath at least one bumper coupled to an inner surface of the upper heel portion, the

at least one bumper configured to secure the upper construction in the closed position; and wherein the sole portion remains substantially planar or does not bend during transition between the open configuration and the closed position.

5. The rapid-entry shoe of claim 4, wherein the at least one bumper is comprised of a foam material.

6. The rapid-entry shoe of claim 4, wherein, in the open configuration, the rear portion of the footbed is angled with respect to a forward portion of the footbed.

7. A rapid-entry shoe comprising: a sole portion; a footbed; and an upper, the upper comprising an upper forward portion and an upper heel portion; wherein the sole portion is coupled to the upper heel portion; wherein the footbed is coupled to the upper forward portion; wherein the footbed is configured to pivot relative to the sole portion between an open configuration, in which at least a portion of the footbed is angled relative to the sole portion and the sole portion retains a substantially flat profile, and a closed position, in which the footbed and the sole portion are substantially parallel to one another.

8. The rapid-entry shoe of claim 7, wherein the upper forward portion comprises a foot hold extending completely under the footbed between opposing sides of the footbed.

9. The rapid-entry shoe of claim 8, wherein the foot hold comprises one or more of a strap and a plurality of laces.

10. The rapid-entry shoe of claim 7, wherein, in the closed position, a rear edge of the footbed is positioned underneath at least one bumper coupled to an inner surface of the upper heel portion, the at least one bumper configured to secure the footbed in the closed position.

11. The rapid-entry shoe of claim 10, wherein the at least one bumper is comprised of a foam material.

12. The rapid-entry shoe of claim 7, wherein, in the open configuration, a rear portion of the footbed is angled with respect to a forward portion of the footbed.

13. The rapid-entry shoe of claim 10, wherein the bumper is spaced vertically above the sole portion.

14. The rapid-entry shoe of claim 10, wherein, when in the closed position, a rear portion of the foot bed is positioned sandwiched between the bumper and the sole portion.

15. The rapid-entry shoe of claim 10, wherein the bumper extends toward a collar of the upper, wherein the bumper flares rearwardly toward the collar of the upper.

16. The rapid-entry shoe of claim 7, further comprising a securement element configured to secure the footbed to the sole portion.

17. The rapid-entry shoe of claim 16, wherein the securement element includes one of a ratchet and a strap.

18. The rapid-entry shoe of claim 17, further comprising a securement release or securement tether.

19. The rapid-entry shoe of claim 18, wherein the securement release or securement tether is configured to interface with the securement element to secure the footbed to the sole portion.

20. The rapid-entry shoe of claim 19, wherein the securement element is positioned on one of the footbed, the sole portion, and the upper.
